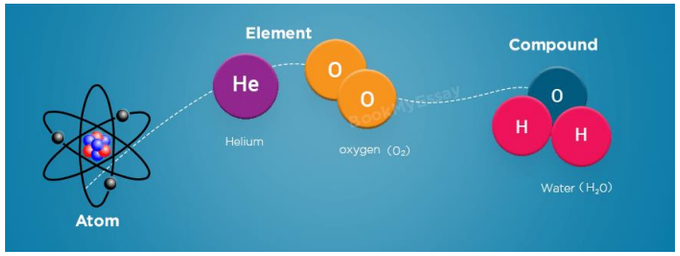
Knowledge Organiser
Year 8: Periodic Table – Atoms and Reactivity

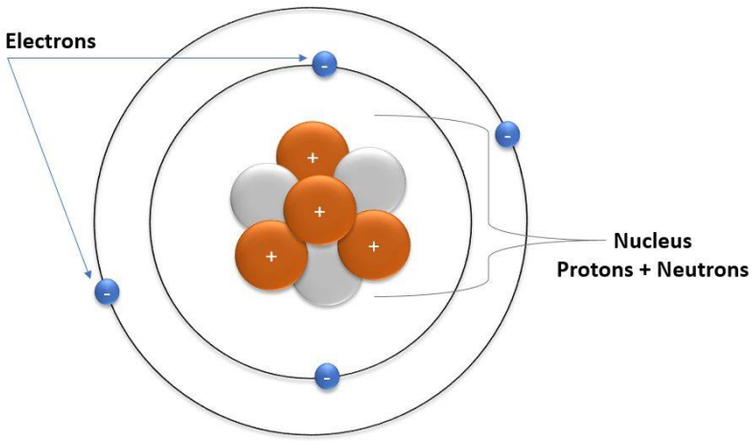
What are atoms and elements?
- Atoms are the simplest units of matter
- If you break down a substance into the smallest bits possible, eventually you will get atoms
- Elements are different types of atoms
- Each element has unique chemical and physical properties

Diagram which compares an atom, element and compound



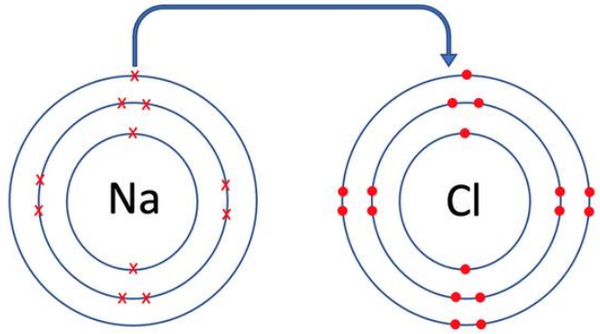
What is the structure of an atom?
- Atoms are made from three subatomic particles
- Most of the mass of the atom is concentrated in a central nucleus
- The nucleus contains the protons and neutrons
- The electrons orbit the central nucleus in shells

Diagram which shows the structure of the atom and the position of the three subatomic particles



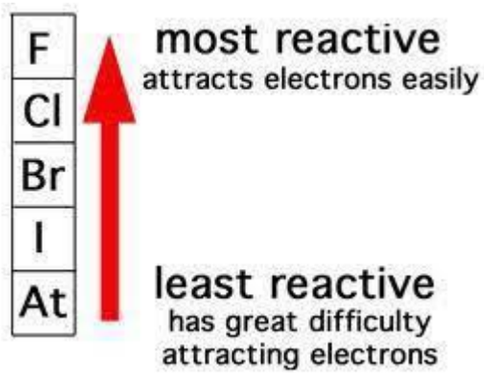
	Charge	Mass	Position
Protons	+1	1	Nucleus
Neutrons	0	1	Nucleus
Electrons	-1	Very small	Orbiting the nucleus

What are the properties of group 7?
- The elements in group 7 of the periodic table are called the **halogens**
- They are in the column labelled 7 at the top
- They have low boiling points
- Their molecules are diatomic (their atoms come in pairs)
- The **halogens** are quite reactive compared to other elements



What happens when they react?
- Group 7 atoms have 7 electrons in their outer shell
- When they react, they **gain an electron** to fill their outer shell to 8
- Most atoms need to gain **more than one** electron to fill their outer shell
- Group 7 are very reactive because it is easy to **gain just one electron**

A diagram which explains the trend in reactivity down group 7



What is a displacement reaction?
- The word displace means **push out**
- A **displacement reaction** occurs when a more reactive element pushes out a less reactive one from its compound

Diagram which shows the displacement of an atom from a compound during a chemical reaction



Using the reactivity series:
- We can use the reactivity series to figure out if a displacement reaction will occur
- If a metal is introduced that is more reactive it will displace the less reactive metal from its compound

Reactants	Products
magnesium + iron sulfate	magnesium sulfate + iron
zinc + silver nitrate	zinc nitrate + silver
gold + copper oxide	No reaction
copper + tin iodide	No reaction
lead + platinum chloride	lead chloride + platinum
sodium + copper phosphate	sodium phosphate + copper

RAG Yourself!

Learning Outcomes;	RAG
Describe how a compound is formed from elements	
Draw a labelled diagram of an atom	
State the charge of the three subatomic particles	
Describe the properties of elements in group 7	
Describe the trend in reactivity down group 7	
Explain why displacement reactions occur	

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What are atoms and elements?

Diagram which compares an atom, element and compound

What is the structure of an atom?

Diagram which shows the structure of the atom and the position of the three subatomic particles

	Charge	Mass	Position
Protons			
Neutrons			
Electrons			

What are the properties of group 7?

What happens when they react?

A diagram which explains the trend in reactivity down group 7

What is a displacement reaction?

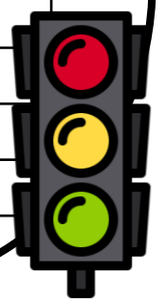
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